

Immersive AI-Powered Language Learning Experience in Virtual Reality: A Gamified Environment for Japanese Learning

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Figure 1: Screenshot of the immersive language learning system, where users interact with AI characters in a simulated environment.

ABSTRACT

This work presents an AI-driven language learning system that combines immersive VR technology and gamification to explore alternative approaches in language education. Developed for Japanese language learners, the system creates a simulated Tokyo environment where learners engage with AI-powered virtual characters to complete tasks, such as purchasing items at convenience stores and seeking directions. It provides dynamic vocabulary assistance and adaptive feedback to support practical communication skills through authentic interactions. The system adjusts task complexity and vocabulary difficulty according to user performance, while reinforcing language acquisition through contextualized conversations. The implementation explores how the integration of artificial intelligence with virtual environments may contribute to computer-assisted language learning.

Index terms: Immersive learning, Gamification, Educational virtual reality, Computer-Assisted Language Learning

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1 INTRODUCTION

Traditional approaches to language acquisition, including classroom instruction, self-study materials, and mobile applications, often encounter limitations in bridging the gap between theoretical knowledge and practical application [1]. While these conventional methods establish linguistic foundations, learners commonly experience difficulties when confronted with unpredictable real-world conversations, potentially impacting their communicative confidence and fluency [2]. Immersive learning experiences, such as study abroad programs and cultural engagement, have shown benefits in language acquisition; however, these opportunities remain limited for many learners due to various constraints, including financial resources, time commitments, and social factors. Even when such opportunities are accessible, learners may face obstacles due to insufficient guidance, limited practice opportunities, and difficulties in unfamiliar social settings.

Recent technological advances in artificial intelligence (AI), virtual reality (VR), and gamification present potential approaches for creating personalized and immersive language learning experiences. Through AI, learners receive contextualized feedback and personalized vocabulary suggestions tailored to their needs [3]. VR facilitates immersive authentic practice while reducing the anxiety of real-world interactions [4], and gamification methods sustain motivation through interactive elements and rewards systems [5]. Previous works and existing platforms [4], such as *Mondly VR*, and *Immerse*, have explored

these elements and demonstrated their potential. However, current solutions often feature structured exercises, fixed scenarios, or limited adaptive capabilities, highlighting the need for more flexible and responsive language learning environments.

Our work attempts to explore these challenges through an integrated approach combining AI, VR, and gamification technologies. The VR component provides an immersive, multi-sensory, and low-stakes environment for practicing Japanese in culturally authentic settings, while gamification elements incorporate task-based activities to encourage learner engagement. AI adapts to learners' progress by offering personalized vocabulary suggestions and contextual feedback to develop practical communication competence.

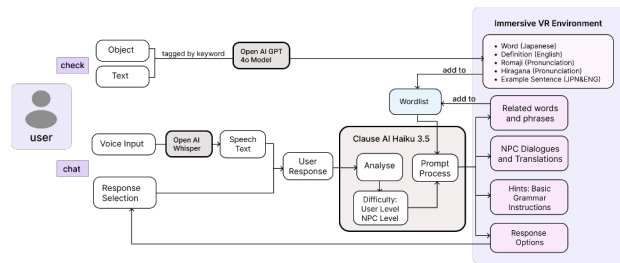


Figure 2: System Architecture

2 METHODS

The system was developed using Unity Editor 2022.3.55 on a PC with an Intel Core i7 processor and NVIDIA RTX 3080 GPU. Virtual scenes were built with assets collected from the Unity Asset Store and Sketchfab, while NPC avatars were created using Ready Player Me platform. The Meta Oculus Quest 3 serves as the VR display device.

2.1 Scene & Object Interactions

The system includes a dynamic dictionary designed to support vocabulary retention by applying spatial memory, which suggests associating information with physical locations and objects may facilitate vocabulary recall [6]. When users click on objects or signs in the VR environment, these items—tagged with specific keywords—trigger requests to OpenAI's API. In response, the AI provides comprehensive language information, including definitions, pronunciations (in both Romaji and Hiragana), and example sentences.

2.2 AI Chats

The AI Chat system is a central component of the adaptive language learning experience, utilizing multiple APIs to generate dynamic and personalized conversations. Claude 3.5 Haiku is implemented for NPC dialogue generation, creating contextually relevant interactions tailored to specific scenarios. OpenAI Whisper enables speech-to-text processing for verbal practice, while Microsoft Azure Speech Studio provides natural-sounding Japanese voice synthesis. The system adjusts dialogue difficulty dynamically based on user performance, analysing grammar, fluency, and contextual relevance. It also offers learning aid features, such as grammar tips, translation assistance, and vocabulary suggestions, supporting the learning process. Users can save vocabulary encountered during interactions, which can be incorporated into subsequent dialogues to reinforce contextual learning and retention. NPCs are configured with specific roles such as shopkeepers or travel guides, to facilitate practice across different conversational scenarios. This setup aims to simulate

natural conversational scenarios, providing structured practice and supportive feedback to facilitate language skills and confidence building.

2.3 Gamifications

Drawing inspiration from games like *The Sims*, the system incorporates gameplay elements to create an interactive language learning experience. A life system introduces mechanics such as hunger and fatigue, encouraging users to engage in daily tasks like buying food or resting at home. A money system requires users to purchase translation assistance, discouraging over-reliance on direct translations while motivating participation in "Work Mode" to earn virtual currency. The map features various daily life scenarios, allowing users to practice different types of interactions - from earning money and practicing customer service language as a café barista in "Work Mode", to engaging in casual conversations in "Date Mode". These gamified features aim to create a more engaging learning environment.

3 CONCLUSION AND FUTURE WORKS

This project proposes an integrated approach to language learning, combining VR environments, game elements, and AI technologies to explore potential solutions for traditional language education challenges.

Future development will focus on comprehensive system evaluation through user testing, examining both technical aspects such as usability and engagement, and learning effectiveness through pre- and post-testing of language proficiency. User feedback on AI interactions and game features will guide system improvements, while performance metrics such as vocabulary retention, conversation fluency will inform design refinements. Additionally, further exploration will explore collaborative learning features, adaptation to other languages, and possible applications in professional and educational contexts.

REFERENCES

- [1] Carmelin P. Mosa, "Language Teaching Methodologies for Effective English Learning," *Int. J. Adv. Res. Sci. Commun. Technol.*, pp. 900–905, Jun. 2023, doi: 10.48175/IJARSCT-11981.
- [2] N. Mediha and M. Enisa, "A Comparative Study on the Effectiveness of Using Traditional and Contextualized Methods for Enhancing Learners' Vocabulary Knowledge in an EFL Classroom," *Procedia - Soc. Behav. Sci.*, vol. 116, pp. 3443–3448, Feb. 2014, doi: 10.1016/j.sbspro.2014.01.780.
- [3] J.-B. Son, N. K. Ružić, and A. Philpott, "Artificial intelligence technologies and applications for language learning and teaching," *J. China Comput.-Assist. Lang. Learn.*, Sep. 2023, doi: 10.1515/jccall-2023-0015.
- [4] T. K. Dhimolea, R. Kaplan-Rakowski, and L. Lin, "A Systematic Review of Research on High-Immersion Virtual Reality for Language Learning," *TechTrends*, vol. 66, no. 5, pp. 810–824, Sep. 2022, doi: 10.1007/s11528-022-00717-w.
- [5] N. Azzouz Boudadi and M. Gutiérrez-Colón, "Effect of Gamification on students' motivation and learning achievement in Second Language Acquisition within higher education: a literature review 2011-2019," *EuroCALL Rev.*, vol. 28, no. 1, p. 40, Mar. 2020, doi: 10.4995/eurocall.2020.12974.
- [6] B. Feng and L.-L. Ng, "The Spatial Influence on Vocabulary Acquisition in an Immersive Virtual Reality-Mediated Learning Environment," *Int. J. Comput.-Assist. Lang. Learn. Teach.*, vol. 14, no. 1, pp. 1–17, Mar. 2024, doi: 10.4018/IJCALLT.339903.